Ines Hipolito: "Dysfunctional Markov Blankets: From Stuck States to Adap

23:27

all right the next video is 23 minutes
long in s hippolito dysfunctional markof
blankets from stuck states to
[Music]
adaptation hi everyone I'll be
presenting today uh dysfunctional Mark
of blankets from Stu states to
adaptation this a paper that has been um
in progress for quite some time and it
is um a collaboration with
maasa Alex Kea and Daniel
fredman so I want to start by uh
highlighting that there are stuck
systems or stuck States all around us
and that we find that example in
neurocognitive um levels such as in OCD
patterns in PTSD trauma Cycles or in
addiction circuits then on the social or
ecological scale uh we find it for
example in poverty traps in eoch chambas
and Market uh monopolies and climate
tipping points then in AI we also find
them for example in a large language
models Loops who isn't been stuck in a I
apologize kind of cycle then uh in
reinforcement learning local Minima in
recommendation filter Bubbles and in
neural networks training fails so what
um these kinds of systems uh share is
that uh they are some sort of critical
threshold uh they have um or present um
self reinforcing loops and uh they have
this persistance to uh change so they
tend to remain in the same state hence
why they are stuck so then the core

problem is going to be a sort of a
paradox so um the view is that systems
adaptive or complex systems um coupled
with the environment they should adapt
but sometimes they don't so then are
from this perspective a few questions um
become Salient why do flexible systems
become so rigid and do not have that
flexibility to adapt to the
environment how do these adaptive
boundaries uh between the system and the
environment break down and what
maintains that dysfunctional State what
is there that must be so strong that um
the system cannot uh change to a
different uh or look for or act for uh
to attain a different more preferable
kind of
state so then of course that these
presents um these are presented as
Challengers because there is an
everchanging world and the system that
needs to adapt to that everchanging
World um and uh it seems like um
existing approaches um such as dynamical
systems theory and network uh science um
still have some limitations to quantify
and address this kind of phenomenon that
we call a dysfunction so then from our
perspective uh we need um a frame work
that unifies that uh unifies this
explanation so we need three things
first we need to identify and
characterize D functional States then we
need to provide a mechanistic
explanation for the Stu States and
finally we need to offer a principled
approach to designing these kinds of

interventions and that's what we set up
to do uh in this paper and that I'll be
presenting to
you so um our framework Bridges um
theoretical Frameworks uh from complex
systems theory and the free energy
principle the main or core um Concepts
uh they are going to be useful are
Notions such as emergence
self-organization and adaptation from
complex systems theory and coupling
self-maintenance and free energy from
the free energy principle by uh
utilizing these two Frameworks uh we
developed uh what a construct that we uh
defined as dysfunctional Mark of
blankets and with that uh we hope that
we will offer a formal theory of uh
systems
dysfunction and we hope to develop
quantifiable measures for this
phenomenon of stuckness and finally
practical recovery strategy so that we
can act upon uh the system that is stuck
in a therapeutic way and we uh uh hope
to kick the system out of that stuck
State into a healthy
State okay so is adaptive systems we all
know that um they are comprised by
components which interact and from that
interaction we observe some sort of
emergence um for example in the case of
an immune response we have different
immune cells each one with a specific
role and they interact in the form of
cell signaling Rec recognition of events
and that interaction then gives rise to
emergence that we observe in the form of

coordinated defense or system level
memory then uh from the free energy
principle which we uh are very
acquainted with uh in this conference uh
we learn that systems that exist act to
minimize their free energy and um
further to that um the understanding or
the Insight is that systems are couple
with the environment and they act upon
the environment to precisely uh attain
or look for more preferable States for
their self-maintenance and we can
explain um that behavior through um
active
inference then another very useful uh
construct is Mark of blankets of course
the framework is a dysfunctional Mark of
blankets and the mark of blanket uh as
we all know is a statistical boundary
between what is inside of the system and
what is outside but further to that uh
an important uh Insight is the
conditional Independence between
internal and external states which give
um the uh system sort of a very
important boundary um to have a
distinction between the self and the no
self and that can be seen also through
the notion of it's an operationally
closed system A system that self
organizes and maintains itself for that
maintenance it's also quite important
that it is thermodynamically open it is
an open system it's not a closed system
and that's what allows them to remain
alive by adjusting to the environment
and here you see for example we have a
neurons in the form of internal states

that could be part of a
network so a key Innovation concept that
we bring in is the notion of
selectability and selectability is a
systems capacity to explore navigate and
select among possible States these can
also be seen as the system having
multistability many points um that are
afforded to the system in the
environment such that uh the system can
be healthy in that
navigation so you can see that for
example that comes represented here in
the form of state A B C or D different
points that the system can explore
within the environment and select the
most preferable ones and um the notion
of selectability allows to um think
about State space accessibility or
affordances the adaptation capacity and
the boundary flexibility to adapt rather
than being rigid and stuck in a
particular
state so then uh within the free energy
principle we can think through that
Paradox it becomes if systems minimize
free energy if they Accord to the
principle why would they get stuck does
the principle not work or is it the
system that is uh not in a stuck state
so then uh what we realize is that um
what is not what is really within the
Paradox is that we cannot explain this
persistent maladaptive States and there
is not really a formal treatment of the
dysfunction so then there is a missing
link between Theory and intervention and
our solution is the dysfunctional Mark

of
blankets there are two types of boundary
dysfunction that we can observe uh in
this systems that are not adjusted
attuned adapt to the environment so they
can come in the form of stock States or
blanket breakdown and stock States I
mentioned this in the first slide and uh
for example in the case of depression
what we observe is rigid kinds of
thought patterns reduced emotional
flexibility impaired Social uh
information processing and shortcuts
bypassing healthy emotional
regulation and in a more General way we
can Define these stck states as the self
reinforcing kind of Cycles being trapped
in deep energy Wells and being resistant
to environmental
perturbation then on the other hand we
have blanket break down and an example
of this is the immune system when the
immune system attacks its own tissue and
what happens here is a very interesting
phenomenon of excessive
permeability um on the boundary of the
blanket right loss of system Integrity
so the self no self-d distinction the
system doesn't know what's self and what
is not self that's the loss of of system
Integrity because of the boundary
dissolution okay so then we arrive at uh
the definition of the dysfunctional Mark
of blankets and this this is a
statistical boundary that um
contrastingly has lost its adaptive
capacity for Effective mediation between
internal and external States and the key

characteristics of um the dysfunctional
Mark of blankets are that the system is
persistent in High free energy despite
adaptation
attempts and what we observe is the
breakdown in information processing and
the loss of effective environmental
coupling I will unpack this further but
let me just situate us um for a moment
so uh we will go from the framework to
practical tools to cross scale
applications that's where we would like
to go uh so to do that we need to First
Define quantifiable measures then um
identify a dysfunction index and finally
the recovery metrics then as to
practical tools we need adaptive
perturbations in order to kick um the
system out of the valley and Recovery
strategy and this on a more therapeutic
kind of site and hopefully we'll be able
to apply these cross across
scales so now looking at the framework
itself the first thing that we need to
do is to um find that dysfunction index
and what that is going to help us do is
to measure and identify and delineate
the dysfunction it will measure the
information flowing through the wrong
channels like a diagnostic kind of tool
that um spots the system bypass and the
higher Dev value the more dysfunction
for example like a cell with a
compromised
membrane then uh we will need also a
recovery metric and this will be a pro a
progress kind of indication indicator
for us so it shows a dysfunction

reduction and the thought here is is
that um we ask ourselves are we fixing
the bypass problem so then uh we will
need uh to uh in order to advance with
that we need the landscape uh reshaping
kind of like bringing out a new
geography of points uh that are afforded
to the system making sure that we don't
get stuck again and hopefully we'll
attain the systems
adaptability where we find the question
can it be a explore can it explore and
respond flexibly to uh the environment
so why are these two measures relevant
well they are they will uh serve the
role of being Universal tools across
scales we will be able to apply them uh
to uh all scales from cells to
ecosystems and they will help us
diagnose and track recovery and guide
our next stage which is the intervention
design
okay so now moving to practical tools
adaptive perturbations and Recovery
strategies uh we have adaptive
perturbations here playing a crucial
role and um an Adaptive pertubation is
going to be a controlled intervention
that temporarily disrupts a system state
to facilitate adaptive change and what
we mean by this or our goal is uh not
simply to push a system out of being
stuck but it is to help it to become
more resilient precisely to make sure
that it doesn't come back to the stuck
State as you can see an illustration
here on the graph below so uh then we
Define the Adaptive perturbation design

as in the first term as traditional
intervention approach where we would
like to nudge the system towards uh
these better States such that it adapts
based on the system's response then with
the middle term we can reshape the
energy landscape have a different sort
of more um flexible uh geography that
makes it harder to get stuck again like
smoothing out the deep valleys and then
finally uh we can promote exploration
and that means help the system uh
discover new affordances new
possibilities hence enhance
adaptability um there are kind of like
two examples of adaptive perturbations
that um I just want to mention and uh
one is by the parameter changes where
for example you can think of that as a
temperature modification in cellular
environments and the other one is the
external force or input an example of
that is trans cranial uh magnetic uh
stimulation
so why is this adaptive perturbation
relevant well it combines three
intervention strategies self adjusting
all parameter parameters will adapt over
time um and it uh includes a balance
between immediate Improvement long-term
perturbation sorry long-term prevention
and future adaptability the Practical
impact of um an Adaptive pupation is
that it will help us guide for Designing
interventions and work across different
scales and prevent the kind of uh so to
speak Band-Aid
Solutions right so then why do ppti

matter in the end well for the stuck States they're going to be very relevant to help us overcome energy barriers um break self reinforc in cycles and enable the system to have more States and the capacity to navigate um and explore uh the the environment and select the states that are going to be much more preferable to the continuation and maintenance and self-maintenance and self organization of the system itself in the case of the blanket breakdown which is this is the case of for example the immune system that has lost uh the capacity to differentiate between what is self and what is no self and attacks um its own tissue then utilizing an Adaptive perturbation uh will be uh useful to restore boundary function to stabilize information flow and to reset System Dynamics and therefore uh repair uh the capacity of the system ultimately to recognize what's self and what is no self what is internal States what is external States okay so uh I will recap uh the journey now uh from the dysfunction to the recovery um so first we need to measure the dysfunction then we need to track the recovery then we will need to design interventions and we do that through the Adaptive uh perturbation to implement the recovery finally we need to monitor that kind of recovery such that we now have um been able to uh get the system to be adapted to the environment uh rather than uh just um in a very short term so key relationships that come up

within this journey include going from
higher dysfunction to Stronger
perturbation that is required going to
better recovery to reduce intervention
strength and going from more
selectability to Greater adaptation
capacity future directions uh where we
need to go is to extend uh this
mathematical formalism to non erotic
systems to multiple time scale Dynamics
and coupled systems
interactions in order to do that we also
need to uh develop measurement tools
such as realtime dysfunction metrics
early warning indicators uh that can
predict a Tipping Point or a point of no
return and then we can apply a
preventive measure and cross scale
validity
tests some of the implemental
implementation challenges that um we can
see um include of course measurement
intervention validation and scaling up
and in order to address them we will
need uh better tools more precise tools
standard tests and infrastructure
the potential impact of uh this
framework um is like I said across
scales uh in healthcare for example we
can look into uh Therapeutics for uh any
sort of disease kind of Cycles uh it
includes uh mental health and aging
patterns in climate action uh we can
apply um this uh framework to tipping
points to as a preventive measure and to
Restoration in AI we can apply to uh
develop more robust AI we can also apply
it to uh make more smoother ethical and

governance uh measures for safety in AI
as well as um can help us design
bioinspired AI such that it's a system
that is uh more adapted uh and hence
more effective finally in Social systems
we can also apply framework for uh
policy design for economical stability
and social
cohesion and um coming to the end uh
call for Action um so uh research
priorities include developing standard
standardized standardized metrics and
build shared data sets create
open-source tools and establish cross
disciplinary teams so please join us in
uh theory development tool creation
application testing and knowledge
sharing conclusion and takeaway so we
went in this journey from uh these
functional systems that we observe that
are stuck in a local minimum and we want
to develop a perturbation in order to
help the system become more resilient
and adapt to the environment out of the
stuck State and enance a full recovery
in the long term so then key takeaways
uh are that we hope that we um have
developed the foundations for a formal
theory of system dysfunction
quantifiable measures for the stuckness
phenomenon and a practical recovery
strategy that can be applied that can be
further in the future applied to
different scales
thank you so much for your time please
do not hesitate to contact myself or any
of the authors uh if you'd like to uh
engage or learn more about uh these work